When Networks Compute:

From Connectivity to Deterministic Execution in 6G

5G Practitioner

ARTPARK

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Why Connectivity is no longer enough?

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Applications depend on timing, coordination, and execution

Example: Remote robot control (industrial or drones)

Networks deliver bits reliably, but not outcomes

Deterministic Compute: computation or decision that is produced within a guaranteed time window

Latency alone does not solve this

Latency tells me how fast data moves, not how predictably computation executes once it arrives.

Something more than connectivity is required

What Modern Applications need?

- Robot motion control
- Industrial process control
- Haptic feedback in XR

Real-time execution

Shared state & coordination

- Multi-robot collaboration
- Cooperative perception between vehicles
- Multi-user XR environments

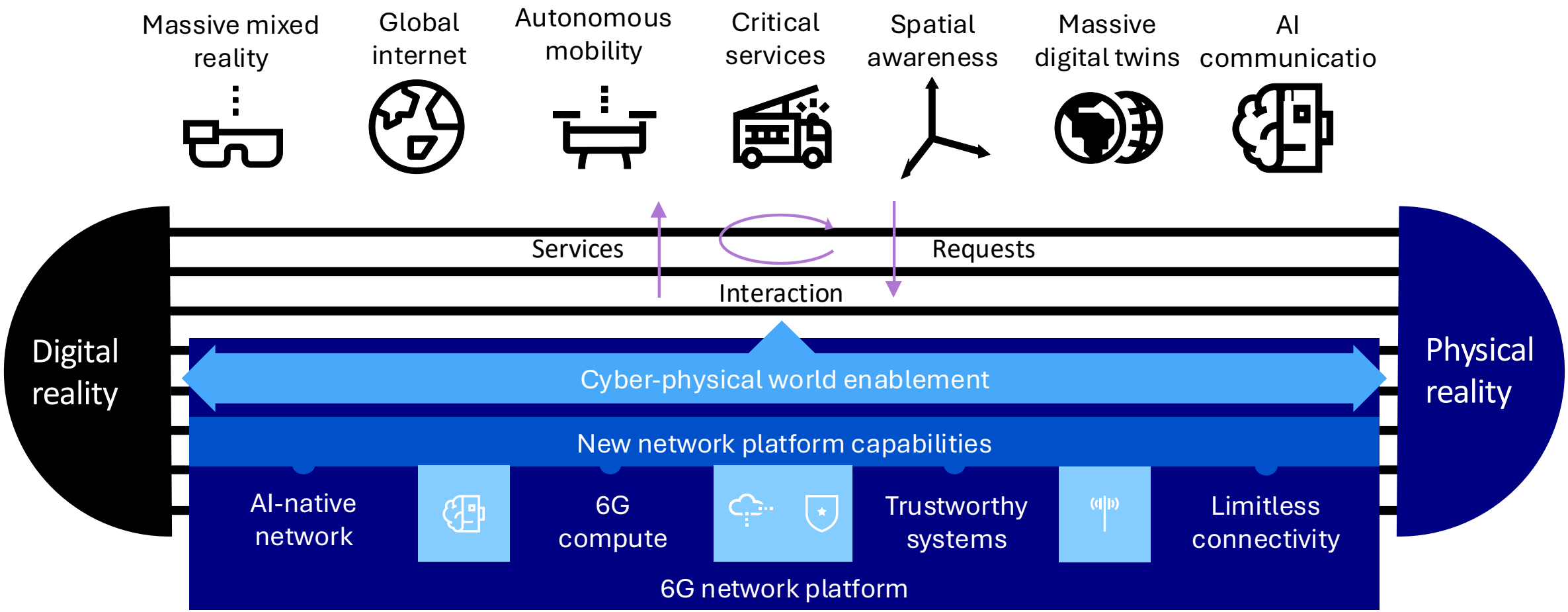
- Drones crossing cells
- XR users walking between coverage zones
- Vehicles switching network conditions

Predictable behavior under mobility

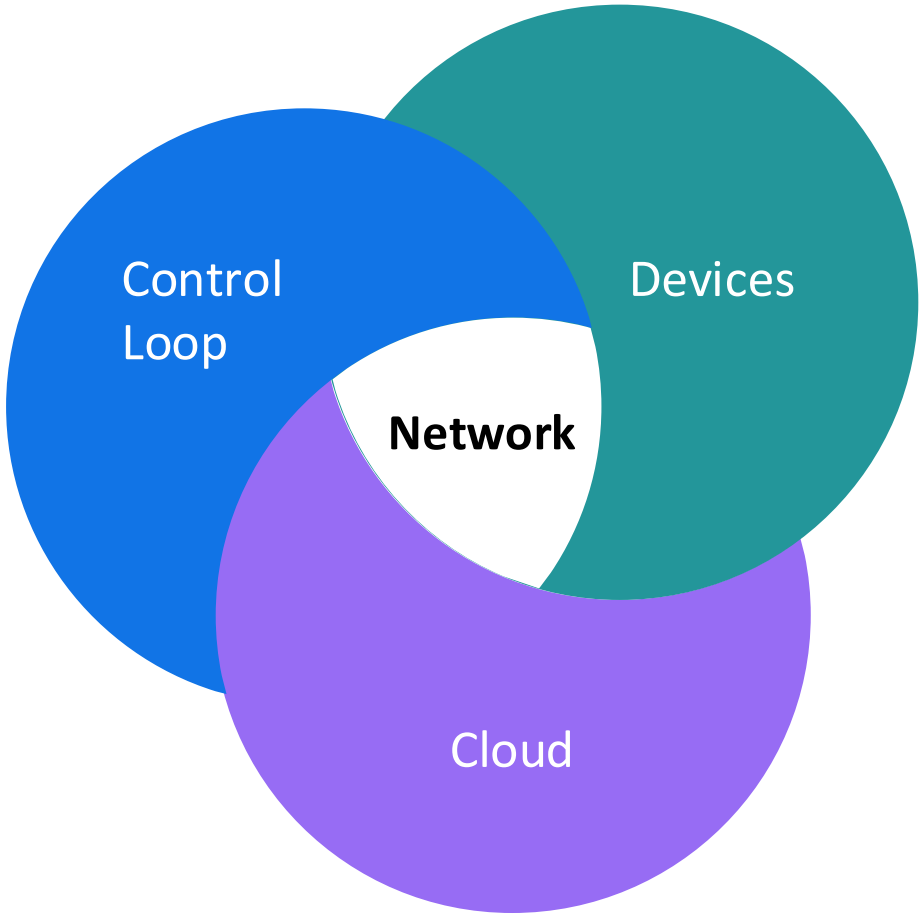
Trust, policy, and locality

- Factory data staying on-prem
- Regulatory data locality
- Safety-certified execution environments

6G: A shift in Network Assumptions



Why Compute enters the Network?



Devices

- Energy and Thermally constrained

Cloud

- Lack mobility awareness & Determinism

Control Loop

- Break across long distance

Example: Robot Control loop

- Sensor data → control logic → motor command
- Loop budget: ~10–20 ms
- Cloud breaks the loop
- RAN / edge sustains it

The Compute Continuum

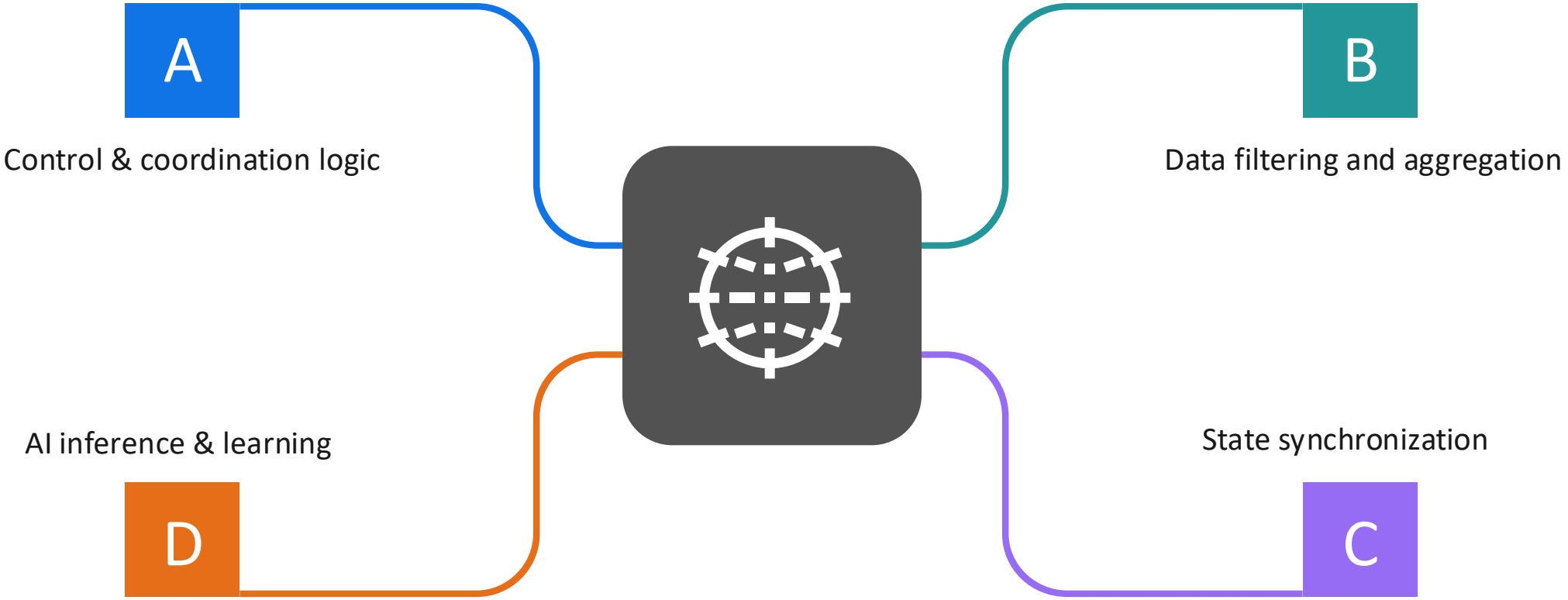
Dynamic placement of execution based on timing, energy, trust, and cost

Device	RAN Edge	Core Edge	Regional Cloud	Central Cloud
Time: sub-	Time: sub-10 ms	Time: 10–30 ms	Time: 30–80 ms	Time: 100+ ms (variable)
Energy: constrained	Energy: shared	Energy: shared	Energy: abundant	Energy: abundant
Trust: personal	Trust: operator	Trust: operator	Trust: operator / regulated	Trust: contractual
Cost: sunk	Cost: premium / bounded	Cost: bounded / pooled	Cost: pooled / predictable	Cost: elastic / usage-based

"Execution and processing move dynamically across this continuum based on timing, energy, trust, and cost
Not an architecture, but a decision space."

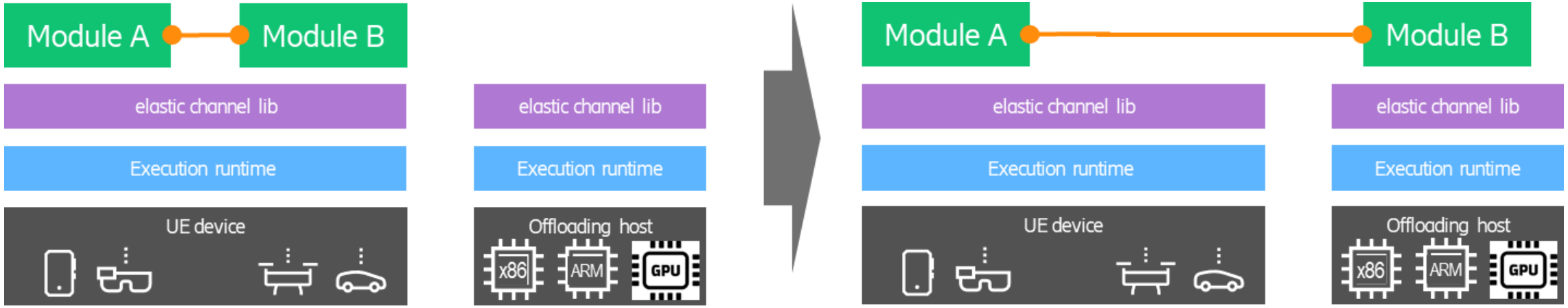
What runs on Network Compute?

Not just AI



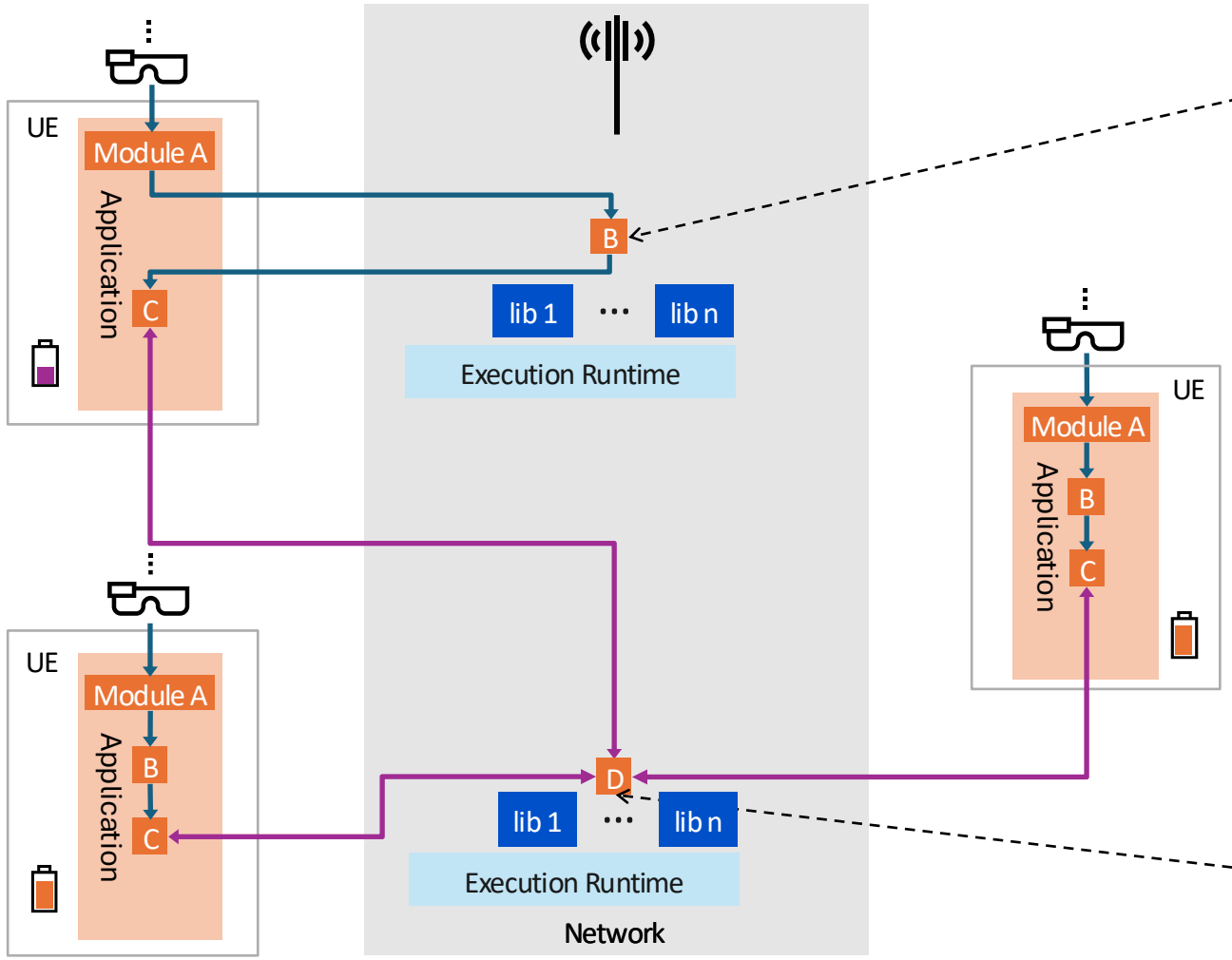
Dynamic device offloading: Background and basic idea

Extending the functionality of mobile devices through dynamic use of remote compute resources



Dynamic Device Offloading

Computational heavy functions can be offloaded to reduce device heat, battery consumption, or computation times



Demonstrated results of this work:

- Use-cases and device agnostic solution
- Dynamic task deployment during runtime
- Supporting platform heterogeneity
- Minimized development overhead
- Quantified benefits for the mobile device
- Network integration with 3GPP EEL and AKMA

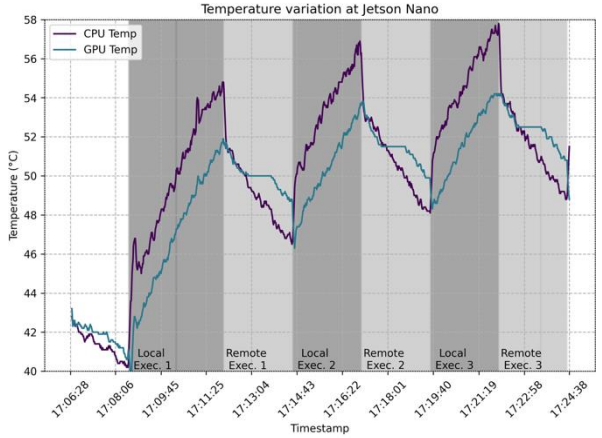
A common coordination task can be offloaded, e.g., to realize collaborative perception or to save overall bandwidth

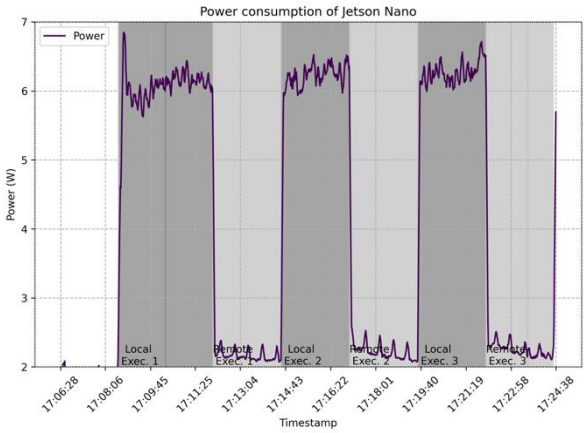
Proof: Why this matters?

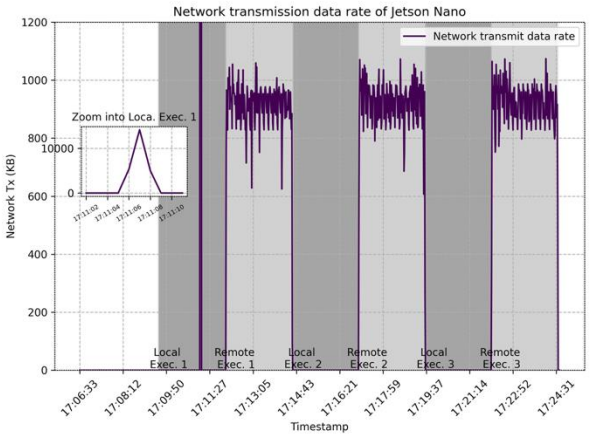
Gains and costs with rest to offloading

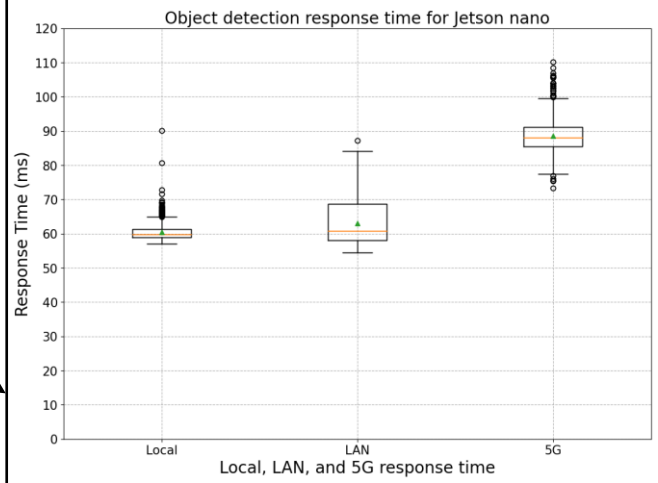
Beneficial
Neutral
Extra Cost

	CPU Usage	GPU Usage	CPU Temp	GPU Temp	Device Power Consumption	Network TX	Network RX	Computational Response Time
Raspberry Pi 3B (limited HW CAP)								
Jetson Nano (moderate to high HW CAP)								









Power vs Performance	
	Energy (mWh per frame)
Local Execution	0.101
Remote Execution-5G	0.058
Remote Execution-LAN	0.050

Computational Offloading Demo

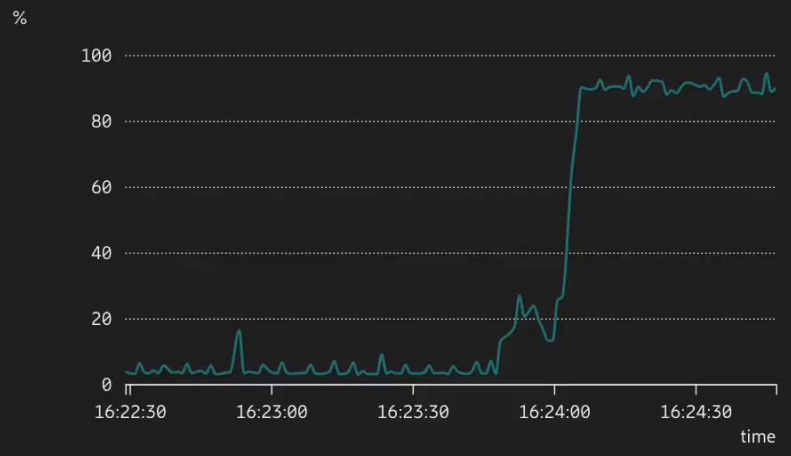
FNP

Offloading Demonstration

Vehicle Object Identification

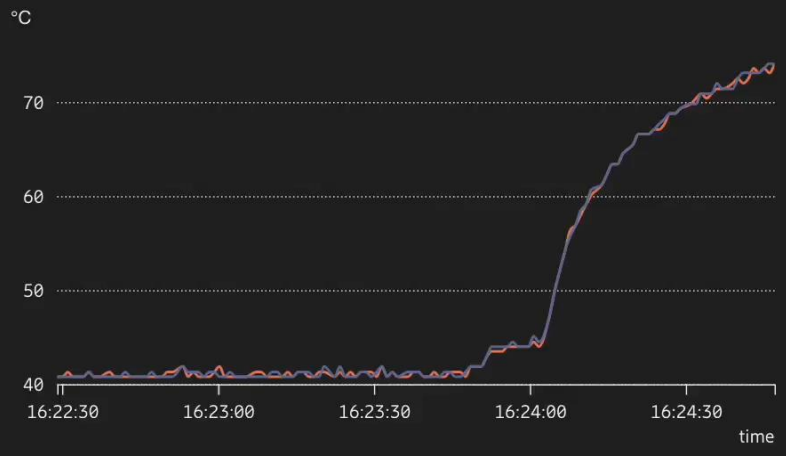
CPU and Network Usage

cpu_usage | network rx | network tx | Select all



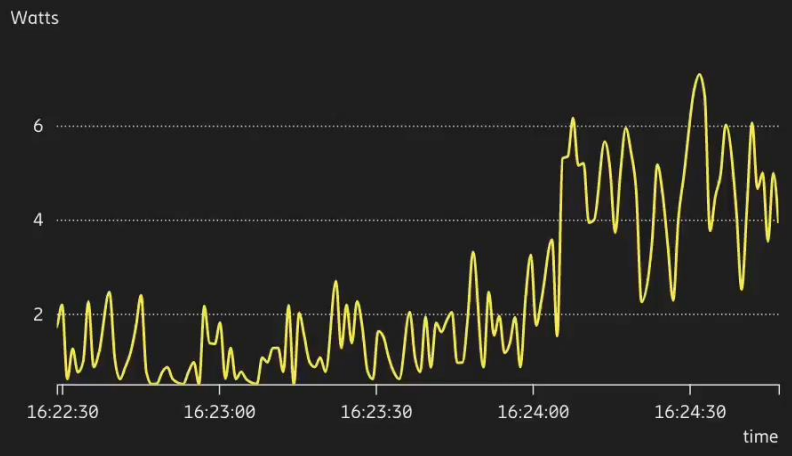
Processor Temperature

temperature cpu | temperature gpu



Power Consumption

power



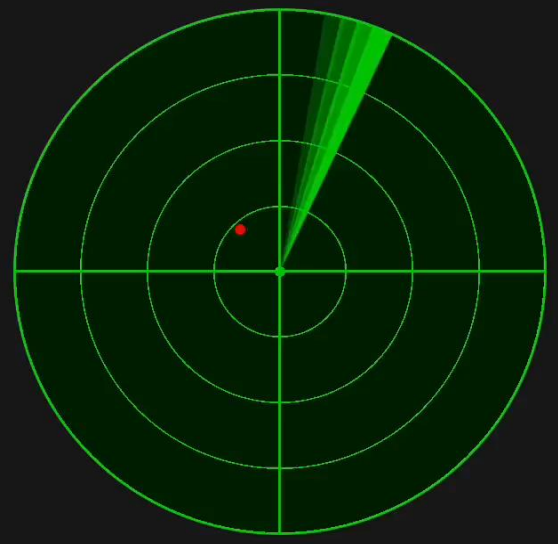
Vehicle Camera Feed Car P

Computational Offloading

Computational Response Time for Object Detection Function [ms]

1363

World View



World Mobile Congress 2024

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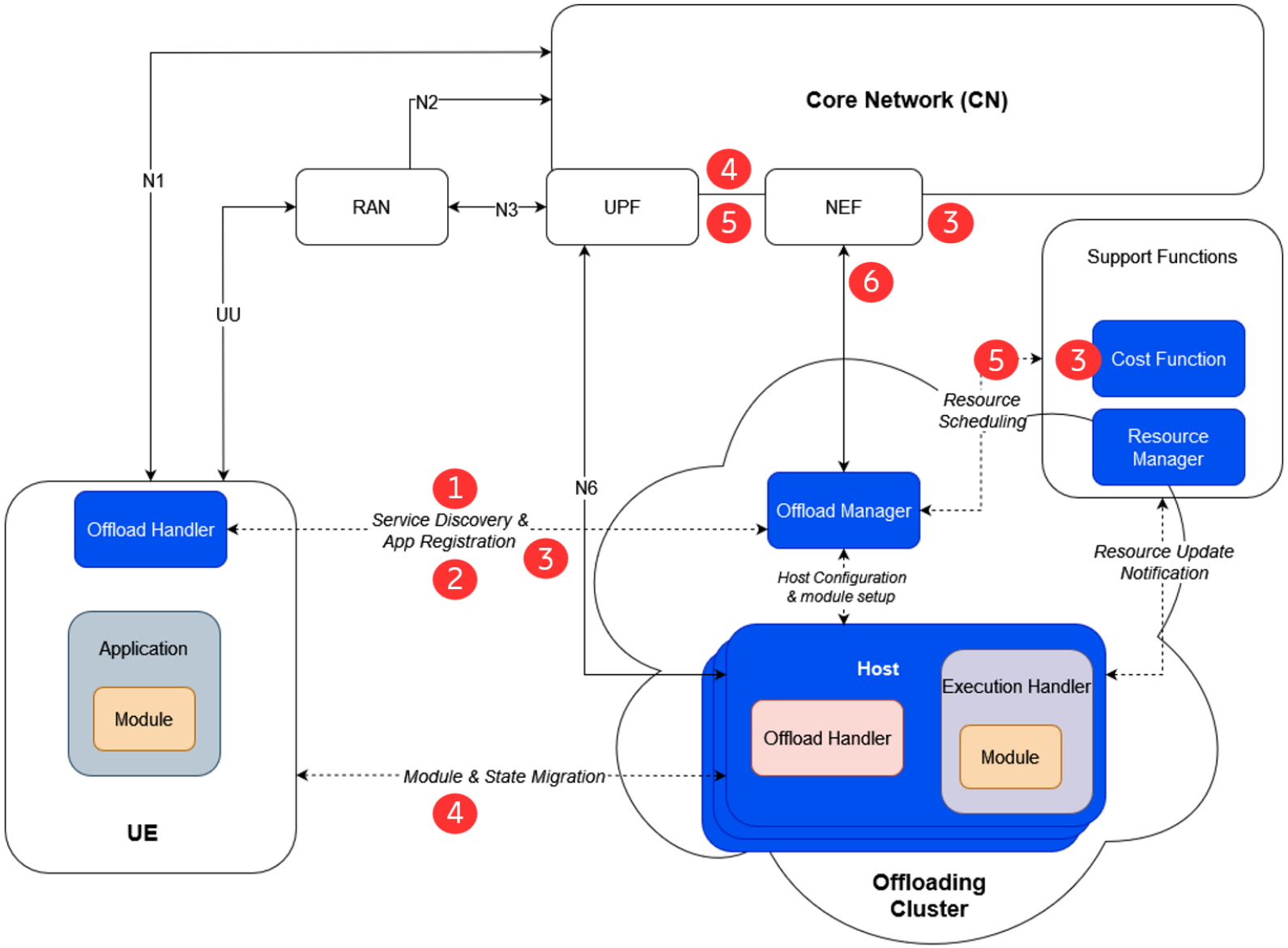
Network Integration Requirements and Architecture draft

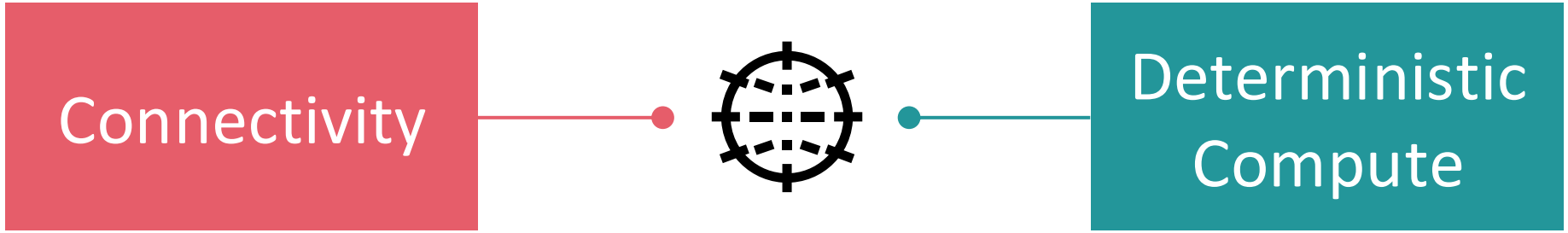
1. Service Registration and Discovery
2. Authentication and Authorization
3. Triggers (based on network info, but also device and remote compute capabilities, and app profile)
4. Setup network userplane
5. Mobility & Appl. Context Reloc. (ACR)
6. Charging and Billing



From Connectivity to Deterministic Execution

Compute Continuum

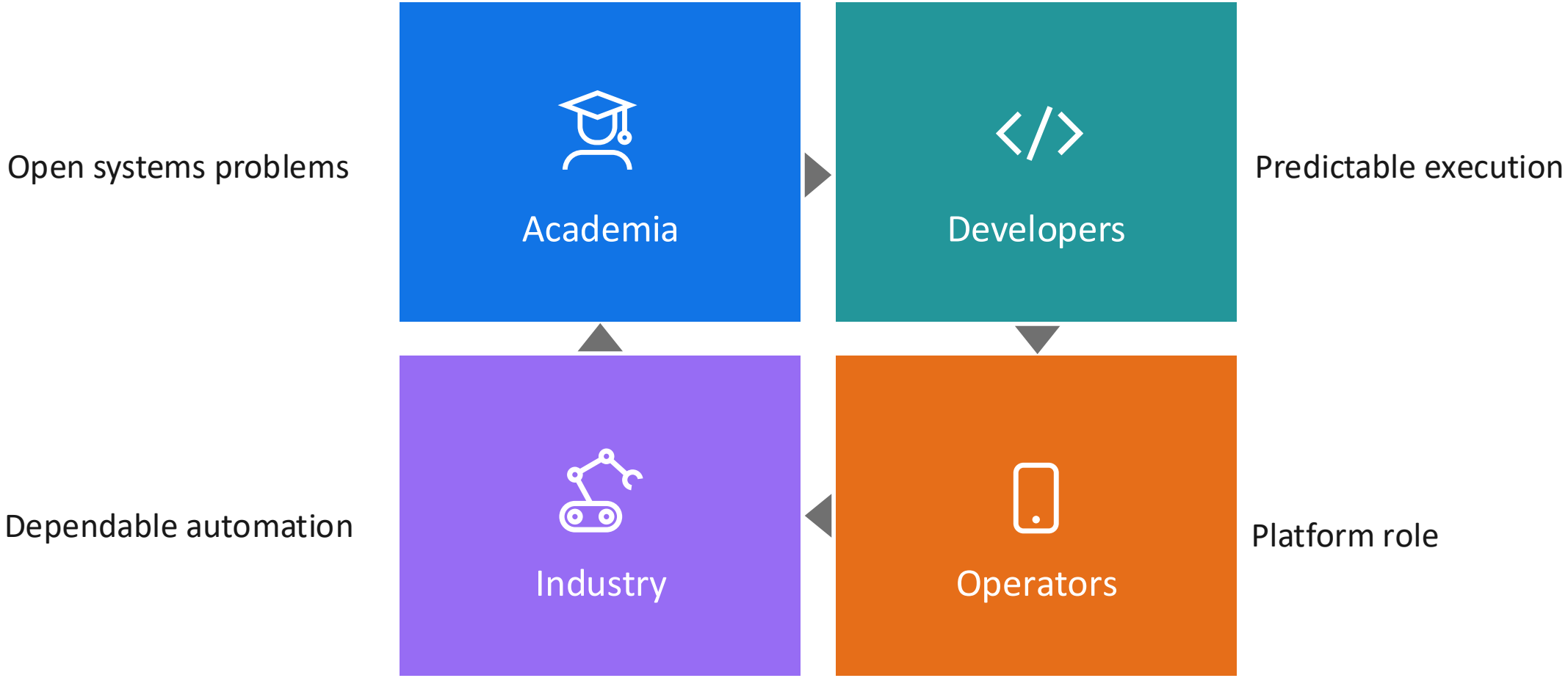
Enables predictable outcomes, not just transport



Moves Data

Executes, coordinates, assures

Who benefits from this shift?



"When networks compute, connectivity becomes capability."

Let’s talk

feedback, comments, questions?

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https://www.ericsson.com/en/edge-computing/network-compute-fabric